

Decent Coaching Classes

VIII SB

Mind Your Math - VII

Answer Key & Marking Scheme

Marks: 40

Time: 1 Hr 30 Min

Q.1 (A) Correct alternatives

[5]

- (1) (B) orthocentre [1]
- (2) (C) $x + 3$ ($x^2 + 5x + 6 = (x + 2)(x + 3)$) [1]
- (3) (B) $(a - b)(a^2 + ab + b^2)$ [1]
- (4) (A) $1/4$ ($k = x/y = 3/12 = 1/4$) [1]
- (5) (D) workers and time to finish a job [1]

Q.1 (B) Solutions

[3]

- (1) Centroid [1]
- (2) $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ [1]
- (3) The time taken becomes half (speed and time are in inverse variation). [1]

Q.2 Any four (2 marks each)

[8]

- (a) Seg LX is perpendicular to side MN (right angle mark), so seg LX is an altitude. [1]
Point Y is the midpoint of side MN (equal tick marks on MY and YN), so seg LY is a median. [1]
- (b) (i) The orthocentre of a right angled triangle is the vertex of the right angle. [1]
(ii) The orthocentre of an obtuse angled triangle lies in the exterior of the triangle. [1]
- (c) Factors of 12 whose sum is 8 are 6 and 2. [$\frac{1}{2}$]
 $x^2 + 8x + 12 = x^2 + 6x + 2x + 12 = x(x + 6) + 2(x + 6)$ [1]
 $= (x + 6)(x + 2)$ [$\frac{1}{2}$]
- (d) Factors of 12 whose sum is -7 are -3 and -4 . [$\frac{1}{2}$]
 $y^2 - 7y + 12 = y^2 - 3y - 4y + 12 = y(y - 3) - 4(y - 3)$ [1]
 $= (y - 3)(y - 4)$ [$\frac{1}{2}$]
- (e) $a \propto 1/b$, so $a \times b = k$. $k = 6 \times 20 = 120$ [1]
 $a \times 10 = 120$, so $a = 12$ [1]
- (f) (i) $p \propto d$ [1]
(ii) $\lambda \propto 1/f$ [1]

Q.3 Any four (3 marks each)

[12]

- (a) Altitude: the perpendicular segment drawn from a vertex of a triangle to the side opposite to it. [1]
Median: the segment joining a vertex of a triangle and the midpoint of the opposite side. [1]
Altitudes are concurrent at the orthocentre (O); medians are concurrent at the centroid (G). [1]
- (b) $2 \times 3 = 6$; factors of 6 whose sum is 7 are 6 and 1. [1]
 $2x^2 + 7x + 3 = 2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3)$ [1]
 $= (x + 3)(2x + 1)$ [1]
- (c) $27p^3 + 8q^3 = (3p)^3 + (2q)^3$ [1]
Using $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with $a = 3p$, $b = 2q$ [1]
 $= (3p + 2q)(9p^2 - 6pq + 4q^2)$ [1]

- (d) $8m^3 - 27n^3 = (2m)^3 - (3n)^3$ [1]
 Using $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ with $a = 2m$, $b = 3n$ [1]
 $= (2m - 3n)(4m^2 + 6mn + 9n^2)$ [1]
- (e) Let cost = x , weight = y . $x \propto y$, so $x = ky$. [1]
 $200 = k \times 5$, so $k = 40$; equation of variation $x = 40y$ [1]
 For $y = 12$: $x = 40 \times 12 = ₹480$ [1]
- (f) Speed (s) and time (t) are in inverse variation: $s \times t = k$. [1]
 $k = 60 \times 4 = 240$ [1]
 $80 \times t = 240$, so $t = 3$ hours [1]

Q.4 Any three (4 marks each)

[12]

- (a) An acute angled $\triangle PQR$ drawn neatly (all angles less than 90°) [1]
 Perpendiculars from P to QR, from Q to PR and from R to PQ drawn using a set-square (three altitudes; 1 mark for two correct, 2 marks for all three) [2]
 The three altitudes meet at one point, named O, which lies inside the triangle. [1]
- (b) An obtuse angled $\triangle STV$ drawn neatly (one angle greater than 90°) [1]
 Midpoints of the three sides found and marked using a ruler/compass [1]
 Each vertex joined to the midpoint of the opposite side (three medians) [1]
 Medians meet at the centroid G, which lies inside the triangle. [1]
- (c) Numerator: $x^2 - 5x + 6 = (x - 2)(x - 3)$ [1]
 Denominator: $x^2 - 4 = (x - 2)(x + 2)$ [1]
 Cancelling the common factor $(x - 2)$ [1]
 $= (x - 3)/(x + 2)$ [1]
- (d) (i) $3x^2 - 12x - 15 = 3(x^2 - 4x - 5)$ [1]
 $= 3(x^2 - 5x + x - 5) = 3[x(x - 5) + 1(x - 5)] = 3(x - 5)(x + 1)$ [1]
 (ii) $27x^3 - 1 = (3x)^3 - (1)^3$ [1]
 $= (3x - 1)(9x^2 + 3x + 1)$ [1]
- (e) More workers means fewer days, so it is inverse variation: $w \propto 1/d$, i.e. $w \times d = k$. [1]
 $k = 20 \times 12 = 240$ [1]
 $w \times 8 = 240$ [1]
 $w = 30$. 30 workers are required. [1]